CubeBot — Fastener Schedule

Hole-by-hole fastener sizes measured from the CubeBot 3D model (`~/cubebot-legstudy/cad/CubeBot_RL/cubebot/urdf/meshes/`)

COMPILED 17 AUGUST 2026 · 7 PARTS EXAMINED · SIZES DERIVED FROM GEOMETRY, NOT FROM A PARTS LIST

The repository contains no BOM. No parts list, no hardware document, and no mention of any fastener designation anywhere in the source. Every size below was measured out of the mesh geometry itself.

There are no nuts in this design. A sweep for hexagonal and square pockets across all seven parts returned zero nut traps. Every fastener threads directly into a plastic boss — specify thread-forming / self-tapping screws, not bolt-and-nut pairs.

1 Top_Case_HD — 125 × 40 × 105 mm · 57,124 triangles · the only fully detailed part

Coordinates are the hole centres in the part's own frame, in millimetres.

HOLE	DEPTH	QTY	FASTENER	LOCATION / NOTE
Ø 1.90 – 2.00	10 mm blind	8	**M2**	Case screw bosses `(±55, ±38.25)` and `(±26.5, ±38.25)`
Ø 2.00	4 mm	2	**M2**	Side wall `(2.6, ±10.5)`
Ø 3.10 `c'bore Ø 3.50`	3 mm `c'bore 2 mm`	3	**M3**	Stepped hole — Ø3.5 lead-in through the wall, Ø3.1 thread-forming pilot behind it. `z = −35.21 / −39.21 / −43.21`
Ø 3.30	9.2 mm	4	**M3**	Enters from the bottom face `(±42, ±27)`
Ø 3.50	5 mm	4	**M3 clearance**	`(32.45 / −25, ±24.5)`
Ø 6.00	3.5 mm	4	*not a fastener*	Bearing seat or cable grommet
Ø 20.00	5 mm	1	*not a fastener*	Connector port `(−60.22, 15, 0)`

2 Remaining parts

Simplified URDF collision meshes, stored in metres and rescaled ×1000 to measure. Holes are cleanly tessellated, so diameters are exact; no counterbores or secondary features are modelled.

PART	HOLE	QTY	FASTENER	NOTE
Bottom_case	Ø 4.25 through 18.2 mm	4	**M4 clearance**	`(±46.5, ±38.5)` — or Ø4 standoffs
Leg_middle_part	Ø 3.00 through 34 mm	2	**M3**	Pivot bolt or 3 mm shaft, `(±24)`
Part_1	Ø 5.00 (2 × 6 mm, 1 × 15 mm)	3	*not a bolt*	Reads as shaft / bearing
Part_1_1	Ø 6.54 × 6 mm	2	*not a bolt*	Bearing seat
Foot	*none*	—	—	No holes of any size
Top_case	*none*	—	—	12-triangle bounding-box placeholder

3 Shopping list

Lengths follow the measured engagement depths. All screws thread into plastic — no nuts required.

M2 × 8–10 mm	×8	Self-tapping, into the 10 mm case bosses
M2 × 4–5 mm	×2	Self-tapping, side wall
M3 × 5–6 mm	×3	Thread-forming, into the Ø3.1 stepped pilots
M3 × 10 mm	×4	Into the Ø3.3 bottom-face bosses
M3 × 6 mm	×4	Through the Ø3.5 clearance holes
M3 × 35–40 mm	×2	Leg pivots, through the 34 mm Ø3.0 bores
M4 × 25 mm +	×4	Through the 18.2 mm bottom case, plus engagement

HOW THESE WERE MEASURED

- Cross-section sweep along all three principal axes at a 0.25 mm pitch. Loops in each slice were nested by containment; a loop at odd nesting depth is an internal void. Round voids were sized by least-squares circle fit through the loop vertices — CAD places tessellation vertices on the true circle, so this recovers the design diameter regardless of mesh coarseness. Consecutive slices were stacked into single features to recover depth.
- Cross-checked with an independent, axis-agnostic pass: faces grouped into smooth-connected patches, each fitted to a cylinder, then classified as a bore or a boss by whether the surface normals face the axis. Both methods agree on every full-circle bore.
- Hex and square nut traps were searched for by corner-counting each void loop (6 corners on a circumscribing circle → hex, across-flats from the enclosed area). None found.
- The OBJ set in `~/cubebot-legstudy/assets/` is byte-for-byte the same geometry as the STL set; measuring either gives the same result.